ADVANCED STATIC ANALYSIS

MALWARE ANALYSIS AND INCIDENT FORENSICS

M.Sc. in Cybersecurity

MALWARE ANALYSIS

M.Sc. in Engineering in Computer Science

A.Y. 2026/2027



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CYBER INTELLIGENCE AND INFORMATION SECURITY

GOALS

Consider the sample provided in the 7z archive (pwd: infected)

- It is not malicious
- You need to understand how it works
- It is a game containing a few puzzles
- You should be able to solve the puzzles

How?

- Reverse the code!

STEP 1

We first need to gain a basic understanding of the target

- Have you tried running it?
- How does it behave?
- Does it expect any input from the command line?
- What happens when you try to interact with it?

STEP 2

Let’s gather some initial information using the basic tools

- Does the PE header tell you anything interesting?
- Strings?
- Imports?

STEP 3

Open the sample in IDA and try to understand its basic structure

- Where is the main() function?
- What does it tell us about the behavior of the sample?
- How is the general control flow organized in the sample?
- Do you recognize code responsible for any of the output you have seen?

STEP 4

Try to overcome challenge #1

- Which function handles the first challenge?
- What does the function check?
- What is the expected input to pass it?

STEP 5

Try to overcome challenge #2

- Which function handles the second challenge?
- How does the function handle input?
- What kind of calculation is performed in the code?
- What is the expected input to pass it?

STEP 6

Try to overcome challenge #3

- Which function handles the third challenge?
- How does the function handle input? Do you recognize any familiar control flow pattern?
- What kind of checks are made on the input?
- What is the expected input to pass it?

STEP 7

Try to overcome challenge #4

- Which function handles the fourth challenge?
- Why does a function call itself?
- Which kind of calculation does the function perform?
- What is the expected input to pass the challenge?

Explore the remaining challenges for further behaviors. Have fun!